



 Final Report
**Pneumonia Detection Using Machine
Learning and Deep Learning**

Epitech Paris

Group 29

1. Introduction

This project focuses on automating **pneumonia detection from chest X-ray images** using both **classical machine learning** and **deep learning** techniques. The goal is to evaluate various approaches, understand their strengths, and build a system that is both **performant** and **interpretable**, a key concern in medical AI.

The dataset is structured into three parts:

- **train/** → used for training models
- **val/** → used initially for tuning, but replaced by cross-validation
- **test/** → held out for final evaluation

Each subset contains two classes:

- **NORMAL** (healthy patients)
- **PNEUMONIA** (infected patients)

2. Dataset Overview

Folder	NORMAL	PNEUMONIA	Total
Train	1341	3875	5216
Val	8	8	16
Test	234	390	624

⚠ Imbalance Warning:

The dataset has many more pneumonia cases than normal, which could lead to biased models. This is mitigated by:

- Using **F1 Score** and **AUC** instead of just accuracy
- Leveraging **class weights** or **resampling** where needed

⚠ Validation Set Note:

Due to the **tiny size of the validation set**, it was not used for tuning. Instead, we employed **cross-validation**, which better stabilizes evaluation and hyperparameter tuning.

3. Preprocessing Pipeline

To make data ML-ready:

-  **Grayscale:** All images were converted to grayscale to simplify computation
-  **Resize:** Resized to **128×128** pixels for uniformity
-  **Normalization:** Pixel values scaled to range **[0,1]**
-  **Structured into:**
 - `X_train, y_train`
 - `X_val, y_val`
 - `X_test, y_test`

For classical ML, images were **flattened** to 1D vectors (16,384 features).

For CNN, they retained their **3D shape**: (128, 128, 1).

4. Core ML Concepts Applied

Concept	Implementation
Overfitting Control	Cross-validation (5-fold), EarlyStopping for CNN
Proper Data Splitting	Used directory split + CV instead of small <code>val/</code>
Evaluation Metrics	Accuracy, F1 Score, AUC
Frugality Awareness	PCA reduced dimensions; reported parameter counts
Interpretability & Scaling	Decision Tree used; discussed 3-class expansion

5. Classical Machine Learning Models

◆ Logistic Regression (Baseline)

- Flattened images to 16,384 features
- Used `GridSearchCV` with `StratifiedKFold(n_splits=5)`
- Scoring metric: **F1 Score**
- Class weights = `'balanced'` to handle imbalance

Best Parameters: `C = 1`

Performance on Test Set:

- Accuracy: 75.5%
- F1 Score: 83.4%
- AUC Score: 90.4%

◆ Decision Tree

- `max_depth=10`
- Simple and interpretable

Performance:

- Accuracy: 71.9%
- F1 Score: 81.0%
- AUC Score: 53.9%

◆ SVM with PCA

To avoid overfitting and accelerate training:

- Used **PCA** to reduce from 16,384 → **100 components**
- Explained **~95%** of variance
- Used linear SVM with $C=1$

Test Performance:

- Accuracy: 73.5%
- F1 Score: 82.3%
- AUC Score: 88.8%

6. PCA – Dimensionality Reduction Justification

Why PCA?

-  **End-to-End Result:** Improved F1 and AUC, reduced training time
-  **Variance Explained:** First 100 components captured ~95% variance
-  **Frugality:** From 16,384 → 100 features → reduced model complexity

Model Complexity Notes

Model	Input Dimensions	Train Time (approx)	Tunable Params
LogisticReg	16,384	Medium	1 (C)
LogReg + PCA	100	Fast	1 (C)
SVM + PCA	100	Fast	1 (C)
CNN	128×128×1	High (~6 min)	100k+

7. Deep Learning – Convolutional Neural Network (CNN)

Architecture

Conv2D(32) → MaxPool → Conv2D(64) → MaxPool → Flatten → Dense(128) → Dropout → Output(2)

- Activation: ReLU
- Output: Softmax (2 classes)
- Optimizer: Adam
- Loss: Categorical Crossentropy
- EarlyStopping (patience=3)

Evaluation (Test Set)

- Accuracy: 80.4%
- F1 Score: 86.3%
- AUC Score: 93.1%

Best performing model overall

 Balanced recall/precision, critical in healthcare

 EarlyStopping controlled overfitting

8. SOM – Self-Organizing Map Visualization

To interpret representations, we used:

- PCA (→ 50 components)
- MiniSom (10×10 grid)
- Color-coded markers:
 -  Circle = NORMAL
 -  Square = PNEUMONIA

Results

- Clear class separation
- Confirms models are learning separable features
- Adds interpretability to PCA+ML pipeline

9. Simulated 3-Class Classification (Bonus)

Although dataset is binary, we explored how to **scale to 3 classes**:

- Modified CNN to output 3 neurons (softmax)
- Adjusted loss to categorical crossentropy
- Used macro F1 and 3×3 confusion matrix
- Suggested datasets: **COVIDx**, **ChestX-ray14**

10. Model Comparison Summary

Model	Accuracy	F1 Score	AUC Score
Logistic Regression	75.5%	83.4%	90.4%
LogReg + PCA	75.0%	83.0%	90.0%
SVM + PCA	73.5%	82.3%	88.8%
Decision Tree	71.9%	81.0%	53.9%
CNN	80.4%	86.3%	93.1%

11. Conclusion

This project demonstrated the full pipeline for AI-based **medical image classification**. We:

- Built strong baselines with **Logistic Regression** and **SVM**
- Applied **PCA** for dimensionality reduction and frugality
- Trained a robust **CNN** showing superior performance
- Visualized high-dimensional features via **SOM**
- Simulated expansion to **multi-class diagnosis**
- Applied cross-validation, tuning, and interpretable methods throughout

✅ The CNN is **recommended for deployment**, while PCA-based models serve as **lightweight, explainable alternatives**, all rooted in sound ML principles.